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RESEARCH-ARTICLE

IoT Games and Gamified Systems: Summertime Sadness or Lust for Hype?

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IoT Games and Gamified Systems: Summertime Sadness or Lust for Hype?

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ABSTRACT

The Internet of Things (IoT) has gained much space in people's everyday lives in the last few years, and it is already possible to see IoT applications widely used in various domains, such as healthcare and industry. The convergence of IoT and digital games also creates opportunities for HCI & Digital Games research, supposedly opening countless possibilities for innovative games and challenging research. However, research on this topic appears to suffer from a generalized lack of clarity in its basics. Following this thread, we conducted a systematic mapping study and analyzed 22 papers that proposed or evaluated IoT games and gamified systems (IoTGGS), aiming to identify characteristics, patterns, and research opportunities related to HCI design and evaluation. However, our own frustrations in this endeavor also led us to identify shortcomings in commonly-used approaches. Thus, this paper contributes to the debate not only by focusing on the convergence between games and IoT with a systematic approach but also discussing the apparently dead-end we have found when closely analyzing this research topic. We present a brief panorama of the field and the identified gaps and raise possible new directions that could help take IoTGGS research to the next level.

CCS CONCEPTS

• **Human-centered computing → Empirical studies in HCI**.

KEYWORDS

Internet of Things, Digital games, IoT games, Gamified systems

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1 INTRODUCTION

The Internet of Things (IoT) has gained much space in people's everyday lives in the last few years. The number of connected things has rapidly increased [3, 53] – projections indicate that, by 2025, the world will have over 75 billion IoT devices connected to the Internet [2] – and soon, IoT technologies will reach almost all aspects of human life [53]. In fact, the spread of such systems has already significantly impacted many sectors of society: IoT applications are widely used in various domains, such as healthcare, industry, automation, and agriculture, to name a few [3, 53].

In this scenario, it is logical to expect that the emergence of IoT would also create opportunities for Human-Computer Interaction (HCI) & Games research and design [42]. For instance, the fact that IoT systems combine digital and physical elements, both augmented with new technological capabilities, opens countless possibilities for research and innovation through games and gamified systems (GS) [17, 53]. Thus, IoT developments have the potential to expand even more the existing games by augmenting the capacity of players' interaction with game devices and transforming the gameplay, opening space for new business models and innovative games and gamified interactions [42].

However, research on such "futuristic and promising" games and GS drawn by IoT[1] appears to suffer from a generalized lack of clarity already in its basics, as we observed in our results and discuss further in this paper. For example, there is confusion on how to name IoTGGS because, besides the "IoT-derived" labels (e.g., *IoT games*, *smart games*, *IoT-based games*, *IoT pervasive games*), the terms *hybrid*, *trans-reality*, *ubiquitous*, *pervasive*, and *location-based* are also indistinctly applied to refer to such applications. Indeed, all these types of games do share traces as they all blur the boundaries of virtual and digital worlds, but they also have differences that often remain unacknowledged in these researches.

Another point of confusion appears when we try to find definitions for what are IoTGGS, their main features, characteristics, or what differentiates them from other types of games – even though there is a great number of scientific papers focused on IoT, it is difficult to find works in the academic literature that offer clear (and at least somewhat precise) direction on this matter. Many papers on this topic limit themselves to presenting their ideas for a game/GS

[1]To achieve more fluid writing, from now on in this paper, we will refer to these "IoT games and gamified systems" by the acronym **IoTGGS**, as it seems logical to relate them directly to IoT in our research.

or reporting technical development efforts without stating their understanding of IoTGGS. In fact, this problem is not uncommon in fields such as HCI, Games research, or Games User Research (GUR). Still, unlike other research topics, apparently, there are no common, structured efforts to progress the field and raise a debate around these issues. A foggy atmosphere seems to revolve around IoTGGS research, which jeopardizes the advancements of research on their interaction styles and how humans can be affected when interacting in this context, either socially, psychologically, or physically.

Such a scenario poses some questions: how are IoTGGS different from their "cousins" (e.g., pervasive, ubiquitous, hybrid, location-based, or AR[2] games)? To what extent is research on IoTGGS exploring the potential and innovation that IoT enthusiasts promote and promise that only IoT can bring? Does it pose specific research challenges regarding the unique characteristics and requirements of users interacting with IoT devices in gaming interaction? Is IoTGGS research just facing a temporary setback, a "summertime sadness," or was its rise nothing but a "lust for hype"? These questions motivated us to conduct this work and raise reflections that (hopefully) encourage new themes, challenges, debates, and future work about interaction, evaluation, and design of IoTGGS.

Therefore, our main objective is to investigate if and how IoTGGS differ from other innovative games that have already been around for a while (i.e., the ones mentioned above). To this end, we conducted a systematic mapping study [32] and analyzed 22 papers that proposed or evaluated IoTGGS to identify characteristics, patterns, and research opportunities related to HCI design and evaluation. However, our own frustrations in this endeavor also led us to identify shortcomings in commonly-used approaches. Thus, this paper contributes to the debate by focusing on the convergence between games and IoT with a systematic approach and discussing with the research and practice community this apparently dead-end we seem to have found when closely analyzing this research topic. We discuss the obtained results bringing a brief panorama of the field and the identified gaps, and we also raise possible new directions that could help take IoTGGS research to the next level.

2 IOT MEETS DIGITAL GAMES

The *Internet of Things* (or simply IoT) can be understood as an ecosystem of common objects equipped with sensors, and network and processing technologies, integrated and working together to create an environment capable of offering intelligent services to end-users [8]. The basic idea of IoT is to surround users with the pervasive presence of various smart things, which can be everyday objects enriched with Radio-Frequency Identification (RFID)[3], tags, sensors, or even smartphones [9, 10]. These objects would then interact with each other and the environment to cooperate and achieve common goals to benefit users.

In the face of that, IoT promises a future of interconnectivity and prospects of new software solutions that could not be achieved otherwise [9, 10]. This promise includes innovative digital games and gamified interactions, posing the idea that through IoT, games and GS could expand their interaction paradigm by using various (smart) devices and sensors [8, 10, 35, 46]. Thus, the popularization of IoT is expected to open new avenues for research and development of these interactive systems, including context-aware games and gamified interactions that more vividly intertwine the virtual and the physical worlds – for example, by incorporating physical objects into the game as interactive elements, positioning players both indoors and outdoors, and sensing players' actions [1, 35, 42].

Some researchers claim that they are already taking advantage of this potential. For example, advances in IoT for smart cities have stimulated the interest of practitioners and researchers of different disciplines in conceiving the city as a playground, promoting an agenda in favor of the *playable or playful cities*[4] [42, 55]. Schouten and colleagues propose creating IoT-empowered games to enhance understanding of complex urban settings [55], while Stokes and co-authors advocate for democratizing game power in smart cities through the Hive Mechanic game engine, enabling the development of purpose-driven IoT games [58]. These initiatives claim to have the potential to address and affect social issues and regional disparities.

Moreover, some researchers have explored combining IoT and gamification techniques for varied purposes. For instance, integrating IoT and gamification has been used to improve user engagement [65], increase patient engagement, and gather data for public health [35]. It has also shown promise in patients' physical rehabilitation [59], enriching museum experiences [38], managing smart homes sustainably [63], and promoting student engagement [25]. These applications leverage the combined power of IoT and gamification to drive positive outcomes in various fields.

Developing such games, however, is not "all fun and games." The literature indicates challenges to developing IoTGGS (e.g., connectivity of game agents, humans, and smart objects interaction), which requires specific design and development skills and HCI knowledge to guarantee a fluid, secure game and offer an engaging player experience [1, 40, 46]. However, if we dig deeper, we can see there are challenges regarding the fundamentals and the conceptual base of IoTGGS before that. As we discuss in the remainder of this paper, the IoTGGS field lacks structured comprehension and oriented approaches that justify and cooperate to advance research on this topic.

For this reason, we share with the reader *our* understanding of IoTGGS to help them in our journey: IoT Games can be understood as a type of game that combines physical objects or spaces with digital technology and connectivity *through the IoT* (the same idea applies to IoT GS). These applications can involve a variety of devices and sensors, such as smartphones, wearable devices, cameras, and environmental sensors, which interact with physical objects and/or environments in hopes of creating a unique and personalized gaming experience. More robust IoTGGS would also use real-world data and events as inputs to the game mechanics aiming to create a highly immersive and interactive experience for players. Besides, some IoTGGS can also include AR and location-based features, for example, but these are additional resources and *not* a strictly necessary or crucial requirement.

[2] Augmented Reality

[3] RFID is a form of wireless communication that incorporates the use of electromagnetic or electrostatic coupling in the radio frequency portion of the electromagnetic spectrum to identify an object, animal, or person uniquely.

[4] This movement fosters a playful urban culture aiming to include games in the citizens' lives and use game principles and mechanics to uncover and understand issues in the urban context.

3 RELATED WORK

There is a considerable and diverse body of literature on systematic studies of the literature around the theme of IoT research. For example, some studies systematically review IoT concepts [3], research techniques and applications [8], technical challenges [35], architectures [52], and benefits and risks of technology adoption [14]. Despite this abundance, the theme of IoTGGS is much less explored, especially when considering the HCI perspective. To the best of our knowledge, no systematic studies have investigated definitions and concepts or the design and HCI evaluation of IoTGGS so far. However, we can find systematic studies for other types of innovative games which share similarities with IoTGGS. We now briefly present some examples, indicating points of contact and contrast with our study.

Kasapakis and Gavalas [29] present a review of the state of the art in pervasive gaming. After following a systematic approach, they offer a comparative view of the generations of these games and provide a comprehensive overview of the different types of pervasive games, the trends in the field, and design principles for creating successful pervasive games. However, their article was published in 2015, and the field of pervasive gaming has evolved significantly since then. As a result, some of the information in the paper is no longer up-to-date. In addition, their review brings a broader focus by including a wide variety of pervasive games.

Carneiro and co-authors [15] present a systematic mapping study focusing on the HCI evaluation of location-based games (LBGs). They conducted their investigation to identify the methods used to evaluate LBGs, the different dimensions evaluated, and the challenges encountered in evaluating the player interaction with LBGs. Although the study provides a clear overview of the topic and indicates practical gaps that permeate the domain, it is limited to a particular subset of pervasive games (i.e., LBGs), different from our subject, and focuses only on interaction evaluation aspects.

Arango-López and colleagues [7] report a systematic literature review (SLR) to identify the methods and metrics used to develop pervasive games and how they relate to Linked Open Data. The authors review the literature on what they call "geolocated pervasive games," analyzing the existing game development methodologies, software metrics, and linked open data approaches used or proposed for such games. While the study identifies the main challenges and opportunities for improving these games and enriching the game content and context, its focus on software and game development methodologies offers more of a Software Engineering perspective.

Last but not least, Paiva and co-authors [47] conducted a systematic mapping on hybrid games in the academy to identify and classify publications on hybrid games in the academic context, offering an overview of development methods, usage scenarios, evaluation methods, and acceptance levels of such games. The advantage of this study is that it analyzes the trends and gaps in the field, suggesting directions for future research; however, it is restricted to the period of 2015 - 2019 and offers limited analysis and discussions of their findings.

Although the mentioned studies do not explore IoTGGS specifically, they offer different perspectives to analyze related games that relate in a broader sense to our object of inquiry and are helpful in our investigation of comparing IoTGGS with other types of games.

4 METHODOLOGY

We performed a systematic mapping (SM) study aiming to construct an overview of IoTGGS and investigate the questions mentioned in Section 1. The SM is a secondary study method that reviews existing primary studies to construct an overview of a research area. Its results may reveal research gaps and trends in one area [32, 49]. Since mapping studies use the same basic methodology as SLRs, we followed the methodology proposed by Kitchenham, Budgen, and Brereton [33], which can be summarized into three main phases: planning, conducting, and reporting the results. We carried out this SM study from September 2021 to May 2022. Below, we briefly explain the main steps of the SM process.

In the **planning phase**, we defined the research protocol to guide the SM process. Our research questions (RQ) encompassed the following dimensions of IoTGGS: **(1) characterization**, **(2) design process**, **(3) HCI evaluation**, and **(4) ethical issues**. We chose these dimensions to gather comprehensive knowledge on the topic and identify research gaps and opportunities related to HCI and games studies. We then formulated the selection criteria for the conducting phase, as shown in Table 1.

Table 1: Selection criteria

Inclusion	The paper presents one or more IoTGGS, reporting the design process OR HCI evaluation
Exclusion	The paper: (i) does not present the design nor HCI evaluation of an IoT game or GS; (ii) is not written in English or Portuguese; (iii) is not a primary study; (iv) has less than 4 pages; (v) is not fully available.

We formed the search string (Figure 1) by using keywords and their synonyms found in papers related to the topic of *IoT games and gamified systems*, along with expert recommendations. We selected these specific terms to focus on IoTGGS and investigate the research community's adherence to the term. As a scope definition, we decided not to build a string to capture studies that focused on other types of innovative games (e.g., hybrid, ubiquitous, location-based, AR, virtual reality) and did not emphasize the IoT element in the game. The authors revised and tested the search string multiple times until reaching the final version.

```
("internet of things" OR iot)
AND
((pervasive OR smart OR digital) AND
(game OR gamification))
AND
(design OR evaluation)
```

Figure 1: String used to search in the sources

We used Microsoft Academic Services (MAS)[5] to find the most relevant publication venues for the research topics *HCI, game design,*

[5]https://docs.microsoft.com/en-us/academic-services/

video game design, video game development, IoT, pervasive game, and ubiquitous computing. After filtering the venues by saliency and period (i.e., last ten years), we identified the top 30 conferences and journals and found the sources that indexed them. Therefore, we conducted our search in five digital libraries: Scopus, ACM Digital Library, Web of Science, IEEE Xplore, and Science Direct.

In the **conducting phase**, we submitted the search string to the sources and obtained 666 papers, including duplicates (Table 2 indicates the number of papers obtained by source). Using the Parsifal[6] tool, we filtered these results in a three-step process to select adequate studies (as illustrated in Figure 2).

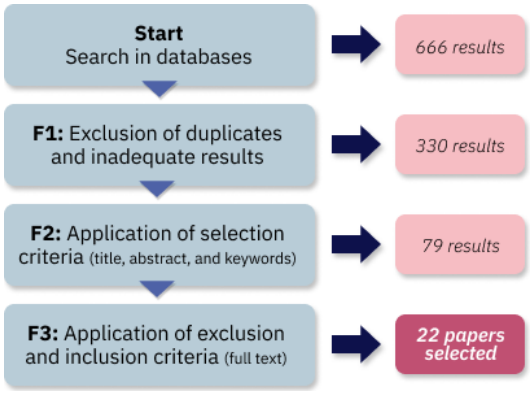


Figure 2: The filtering process followed in the SM

Table 2: Search results and selected articles by source

Source	Nº papers found	Nº papers accepted
Scopus	311	8
IEEE Xplore	190	3
Web of Science	122	9
Science Direct	29	0
ACM DL	14	2

The *first filter (F1)* consisted of removing duplicates and works that fit general exclusion criteria (i.e., secondary or tertiary studies, papers with less than four pages, not written in English, or unavailable). Parsifal automatically removed 187 duplicates (28.07%), and we manually excluded 149 (22.37%), resulting in 330 papers (acceptance of 49.55%). Four evaluators performed the *second filter (F2)* by analyzing each paper's title, abstract, and keywords and applying the selection criteria – F2 resulted in 79 (11.86%) papers.

We tested the consistency of the filtering decisions by having two evaluators analyze a random 20% sample (68 items) of our studies using the selection criteria. We computed the Fleiss' kappa for this inter-rater reliability analysis and obtained a good result (κ = 0.610; 80.9% of agreement), so we continued the filtering. Then, three evaluators performed the *third filter (F3)*, examining the full text of the 79 papers and reviewing the assigned inclusion and exclusion criteria. As a result, we selected 22 papers that presented one or more IoTGGS, namely: [11, 12, 16, 20, 21, 23, 25, 26, 30, 31, 36, 38–40, 44, 48, 50, 51, 59, 60, 63, 64]. It is worth noticing that many papers were excluded for not using IoT as a central tool to develop their IoTGGS or not indicating that clearly in the text. We are aware that this decision may have excluded relevant works, but that was a scope decision, as explained earlier in this section.

Concluding this phase, three researchers extracted and organized the data from the selected works in spreadsheets to perform a quali-quantitative analysis and answer the RQs. The quantitative analysis was mainly exploratory and used descriptive statistics to describe the data relevant to our goals. The qualitative analysis followed a thematic approach [27] guided by our RQs' themes (i.e., IoTGGS characteristics, design process, HCI evaluation, ethical issues, and challenges or difficulties), focusing on the key topics to map and summarize unstructured data in thematic categories. For the IoTGGS characterization, we also mapped their traces using some dimensions of Sethi and Sarangi's taxonomy for IoT technologies [56], supplemented by emergent categories when necessary. We finished the third phase of the SM by reporting the results in the next section.

5 RESULTS

We observed that the selected studies were published between 2014 and 2022 (Figure 3), with 54.5% of the articles published between 2018 and 2020 (n = 12). There is not a clear tendency of evolution in the distribution of publications per year – although there are peaks in 2018 (n = 5; 22.72%) and 2020 (n = 6; 27.3%), they are followed by a decrease in 2021 (n = 1). This scenario requires further investigation, but it is possible that the COVID-19 pandemic affected the research interest in this topic and the evolution of works in progress back in 2020 (the year with more published papers on the subject). We also observed the publication vehicles and identified twenty venues: seven journals and 13 academic events. The papers are distributed in various venues, with no "preferred" publishing location.

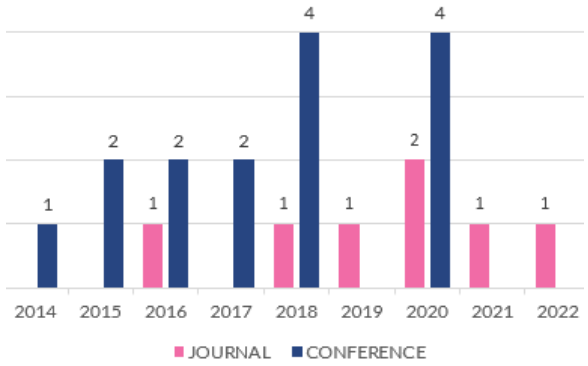


Figure 3: Studies published by year and publication venue

Regarding the geographical distribution of the authors of the selected studies, we identified 16 countries of origin[7] distributed in four continents (America, Europe, Oceania, and Asia). Most studies (20 papers – 90,91%) are from Europe: the United Kingdom leading

[6]https://parsif.al/about/

[7]Countries refer to the place where the authors' institutions are based.

(5 studies), followed by Germany (3). There is a lower incidence of studies in America – only two authors represent the American continent, one from Brazil [40] and the other from Canada [36] – and Africa had no representation in our results. Regarding authors' affiliation, the studies relate to 37 different institutions, being 29 universities or higher education institutes, and eight industry-related organizations.

5.1 IoTGGS Characteristics

The 22 studies present 25 IoTGGS (paper [21] presents four games): 16 (64%) games and six (24%) GS: [20, 36, 38, 39, 48, 63]. We analyzed the identified IoTGGS and classified them regarding the domain of application and purpose, genre (when game), target audience, development status, utilized materials, and the IoT technical elements they employ – Table 4 summarizes the identified IoTGGS[8]. As noted by Sethi and colleagues, standardization is difficult when dealing with IoT systems because of the varied requirements of different applications and devices [56]. This pattern seems to be reflected in most of the analyzed papers, as they provide varying types and levels of information when presenting the IoTGGS, thus hindering the extraction of equally deep data.

Regarding the **domains of application and purposes** (Figure 4), although IoT potential allows the development of applications covering many domains [9], the majority of the found IoTGGS fell into the specific domain of *Entertainment* since the most explored application domain for IoTGGS was *Social life & Entertainment* (10 IoTGGS), followed by *Smart Environment* (4 IoTGGS) and *Energy Conservation* (3 IoTGGS). It indicates opportunities in less explored domains, such as Health and Fitness, Smart Agriculture, and Smart Transport. Besides, the most frequent purposes were Education (9 IoTGGS), Entertainment (8), and Behavior Changing (8).

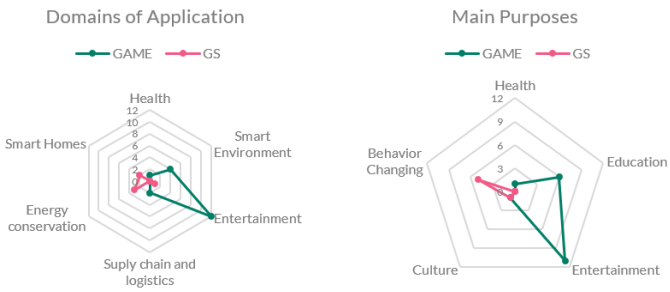


Figure 4: Distribution of IoTGGS domains and main purposes

As for the **target audiences**, most IoTGGS target the general public (n=13, being five GS and eight games). When directed to a specific audience, they were: students (4 IoTGGS), children (3), teenagers (1), families and roommates (1), and – surprisingly – gorillas (1).

The analyzed papers showed IoTGGS in different **development statuses**. Most papers (n = 15) present prototypes to demonstrate the feasibility of their proposals, being five of them focused on GS [20, 36, 39, 48, 63], and ten focused on games [11, 12, 16, 21, 23, 31, 44, 59, 60, 64]. Only five studies present launched IoTGGS: [25, 26, 30, 38, 51]. Two papers focus on the concept of their games [40, 50].

As expected when analyzing games, the authors of the selected papers classified their **game's genre and types** with different criteria – that is a common thing once there are several classification schemes to categorize digital games based on completely different aspects [24]. For example, we find *serious games, scavenger hunts, multi-player online games, and simulations* between authors' classifications. Thus, we classified the games according to their mechanics and assigned one or more genres based on Heintz and Law's classification [24], which recommends five game genres (mini-game, action, adventure, role-play, and resource) and provides a set of defining attributes to help the genre identification. We observed the main traces of the analyzed games' mechanics to assign their genres, as our goal was not to make a definitive classification but to provide an overview. Thus, the most frequent genre identified was Mini-game (6 games), followed by Adventure (5) and Action (3), emphasizing the game structure, players' actions, and goals.

We identified smartphones as the most common **interaction device** used in 15 (68.18%) studies, mainly as a means of end-user interaction. Other materials included Arduino devices, domestic appliances, and plywood. Three studies stood out for using less common devices [21, 23, 44]: [21] used a technology called *BBC micro:bit*, a platform that uses a 2.4GHz radio module that allows simple local networks to be programmed – chosen to protect the privacy of the targeted end-users (children); [44] used projectors and touchable tables to interact with the system; and [23] used infrared and vibration sensors to detect information during the gorillas' interaction with the game.

Considering a **technical perspective** to characterize how the analyzed studies incorporate IoT technologies, we classified them using Sethi and Sarangi's taxonomy for IoT technologies [56], which comprehensively describes IoT architectural elements. We focused on the components of perception (sensors), communication (networks), and middleware layers. In our set of studies, the perception layer commonly featured temperature, humidity, and motion sensors, as observed in six (27.27%), four (18.18%), and four (18.18%) studies, respectively. The most common communication strategy was wireless sensor networks (16 studies), followed by low-power technologies (five studies). Regarding the middleware layer, seven (31,8%) studies opted for application-specific middleware, seven (31,8%) opted for service-oriented, and seven (31,8%) studies did not specify their choices.

5.2 Design Process

In general, the studies focus on three main components of the IoT service structure when they describe their IoTGGS design process: (i) sensors and network technology, (ii) middleware, and (iii) application software technology [23, 46] – we do not discuss these technical aspects here as they are out of our scope of interest. All the 22 analyzed papers described the devices, sensors, and connectivity features employed on some level. Among them, the level of detail provided varies. Some explain a complex architecture with several agents [31, 59], others explain networking strategies [16, 30], and all of them describe diverse types of sensors and data collected or produced. Others even mention even the switch buttons and

[8]Also available in http://tiny.cc/IoTGGS.

the electronic components used in their designs [60]. However, we observed that fourteen (63.64%) papers did not describe the design process, focusing only on presenting the concept/idea of the game in general lines and giving an overview of the infrastructure behind the game – six studies only describe the IoTGGS concept [21, 36, 40, 50, 59, 60].

Analyzing the other eight studies, we observe that six (27.27%) of them partially describe how the IoTGGS were designed [12, 16, 20, 21, 26, 40]. They provide only superficial information related to the design process – for example, mentioning a methodology (e.g., research through design [21]) or referring to an initial data collection to elicit requirements or to prototype (e.g., [16, 26]). Two (9.1%) studies stood out for detailing their processes: [31], which presents a *biotic game*, and [23], a game for gorillas.

We also observed the challenges authors faced in designing and implementing IoTGGS – they were mainly related to hardware and technical problems, the ideation phase, and the physical structure to implement the proposed solutions. Seven (31.81%) of the analyzed studies proposed tools to help others to develop IoTGGS [12, 16, 20, 25, 26, 30, 40]. Notwithstanding, most studies did not communicate concerns in considering human factors when designing their solutions, nor reported following human-centered design principles, for example. Besides, these studies do not register particular challenges related to IoTGGS interaction design. It seems that most authors use their papers to promote a new idea to apply IoT or a technical improvement made on one or more IoT layers.

5.3 HCI Evaluation

Regarding the HCI evaluation, only 11 (50%) papers reported conducting some type of evaluation, while the other half did not evaluate the interaction or experience of their IoTGGS. Looking closely at the first group, we identified that nine studies followed a mixed-methods approach, aiming for quali-quantitative analysis [12, 25, 26, 30, 39, 44, 48, 51, 63]. Two studies followed different paths, one focusing on a quantitative approach [59] and the other on a qualitative one [23].

The results were diverse when observing the **interaction qualities under evaluation**: usability was the most evaluated quality explored in four studies (18.18%) [12, 30, 39, 63]. Other three (13.63%) studies, although they did not refer to usability, reported evaluating qualities commonly included in usability definitions, such as satisfaction, usefulness, and efficacy [23, 44, 51]. Engagement and enjoyment – two important qualities when considering games and gamified interactions – were evaluated in two (9.1%) studies each [25, 26], and only one (4.54%) study [30] evaluated player experience – a frequently explored quality of interaction in GUR.

We also identified the **methods and instruments** the studies used to evaluate the IoTGGS (see Table 3). The three most used were: surveys (including questionnaires and scales), interviews, and user observation. Eight (36.36%) studies applied surveys [12, 25, 26, 30, 44, 48, 51, 63]: seven (31.82%) works report using *ad hoc* questionnaires, and two (9.1%) used questionnaires proposed in the literature [48, 51]. Five (22.72%) studies used user interviews [26, 30, 39, 44, 48], and three (13.63%) conducted user observations [23, 26, 39]. One paper [59] did not mention how the authors performed the evaluation or any method used. Figure 5 relates

Table 3: Evaluation qualities and methods used in the studies

Ref.	Type	Evaluation Approach	Quality Evaluated	Methods
[51]	Game	Mixed	Usefulness, satisfaction, ease of learning and use	Questionnaire (USE)
[26]	Game	Mixed	Engagement and enjoyment	Questionnaire, interview, user observations, video
[25]	Game	Mixed	Student engagement	A/B testing, Ad hoc questionnaire
[30]	Game	Mixed	Player experience, usability	Questionnaire, interview, log data
[44]	Game	Mixed	Satisfaction of navigation, functionality and UI interaction	Survey, interview
[59]	Game	Quanti.	Not specified	Not informed
[39]	GS	Mixed	Adaptability, Usability (efficacy)	Lab experiment observation, Real-life scenario observation, Interview
[63]	GS	Mixed	Usability	Ad hoc questionnaire
[23]	Game	Quali.	Efficacy	Pure observation
[12]	Game	Mixed	User acceptance, usefulness, affective satisfaction, usability	Scale based on [61]
[48]	GS	Mixed	Enjoyment	Questionnaire (PES), Interview

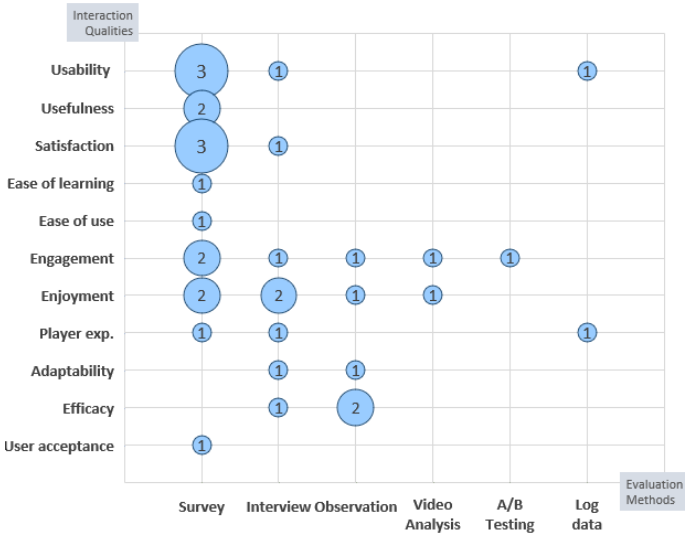


Figure 5: Map relating interaction qualities under evaluation and methods used in the studies

the interaction qualities under evaluation and the data collection methods used in the analyzed studies, in which we can see that the preferred combination when evaluating IoTGGS interaction is using surveys to evaluate usability and satisfaction, even though satisfaction is a component of usability. It is worth noting that the

analyzed papers also did not mention challenges or difficulties related to the evaluation process nor showed particular interest in investigating how IoT impacted user interaction and experience.

5.4 Ethical Issues Regarding IoTGGS

Considering the analyzed studies, the discussion of ethical issues related to HCI and technology adoption was generally absent, and the papers did not present or discuss ethical concerns or difficulties specific to IoTGGS faced during their conduction of the studies. Once the IoT paradigm brings a plethora of ethical issues yet to be studied and dealt with by researchers and practitioners, regarding, for example, privacy, security, and informed consent – and the discussion of this topic is ascending [54, 62] – we included such aspects in our investigation, trying to identify specific ethical issues related to IoTGGS design and evaluation processes reported by the selected studies. However, it was surprising that 18 (81.82%) studies did not mention any concerns or measures focused on such issues. Only four (18.18%) studies mention ethical concerns, namely [23, 25, 26, 31]. Considering these works, the concerns expressed were: consent to participate and allow video recording [26], ethical committee approval [23, 25], respecting ethical issues when testing animals [23], and specific precautions related to Interactive Biotechnology ethics [31]. Even though these studies mentioned the ethical issues above, most papers did not explain or discuss them or their significance and implications. Therefore, we did not obtain information on ethical challenges specific to IoTGGS.

6 DISCUSSION

We now discuss our findings to answer the questions that motivated this study, as posed in Section 1. In subsection 6.4, we present insights regarding future research on IoTGGS.

6.1 Are IoTGGS different from other games?

One cannot help but compare IoTGGS with other types of games that also blur the virtual and digital worlds, especially the umbrella categories of pervasive games. Pervasive games are a type of game that integrates elements of the real world into its gameplay [6]. They use both real and virtual environments, influencing the player experience [6]. Pervasive games present themselves in different subtypes, such as smart toys, augmented tabletop, exergames, AR games, location-based games, and serious games [6, 41], and are also used for a multitude of purposes. Another type of game commonly confused with IoTGGS is the hybrid games family, born from the convergence of digital and analog games [47]. Hybrid games use physical and virtual components to enrich the gaming experience with varying levels of hybridization, from almost entirely analog to almost fully digital [34].

It is clear that IoTGGS, pervasive games, and hybrid games share common features. They all strive to integrate real-world and digital elements, blurring the boundaries between physical and virtual realms. Their appeal lies in taking advantage of technology to offer more immersive experiences. Additionally, the three categories utilize elements of the player's physical environment as a crucial aspect of gameplay. This integration of real-world context aims to enhance engagement and provides a unique experience for players. In spite of that, the answer to our question above is: **yes, there is literature-based evidence that they are different.** Despite their similarities, we can trace what appears to be the main (or maybe only) difference between IoT games, pervasive games, and hybrid games regarding their design and implementation: **IoT games primarily focus on leveraging the IoT technology, wherein physical objects and digital devices are interconnected to enable the gameplay mechanics**. In contrast, pervasive and hybrid games encompass a broader scope and can incorporate any or multiple technologies beyond IoT (such as GPS, RFID, and AR) to create gameplay environments. IoTGGS tend to place significant emphasis on integrating physical objects *into* gameplay, utilizing IoT devices as integral components.

Our findings support this perspective and suggest a predominant technical perspective among researchers investigating IoTGGS. The analyzed studies emphasize the technical aspects of the devices used rather than innovations in the interaction itself and frame IoT as a means to achieve specific objectives. Consequently, a significant portion of the articles lacks relevant discussions on user interaction, human factors in game design, or design and evaluation processes of interaction and user experience in these games. They generally emphasize demonstrating IoT technology's viability, proposing technical improvements, or presenting novel ideas for applying IoT. This may be the main contrast between research in IoTGGS and research in other types of games and gamified interactions, such as pervasive or hybrid games, where there seems to be a greater concern for conceptual issues, human factors, and design interaction. The statement made by Petrovic and colleagues exemplifies this distinction: "Instead of being an object of study, IoT can be used to enhance other fields of learning by providing devices and technology to the classroom" [51].

From an HCI perspective, the distinction between IoTGGS and pervasive games is subtle. In the end, IoTGGS are essentially a subtype of pervasive games. In most cases, the key difference is that a technological "bias" distinguishes IoTGGS, often driven by creators' technical affinity or the need to fit in a hot, trending topic – justifying the IoT label. That may be why conceptual discussions in this field have remained less developed. While that is not exactly a problem, there are some potential drawbacks of an excessive focus on technical viability in IoTGGS research at the expense of conceptual issues and concerns for human factors. It may lead, for example, to a lack of deep conceptual discussions that solidify the area, limited or poor user engagement, and neglect of human-centered design and robust HCI evaluation practices and tools. Perhaps more importantly, it can result in a poor gaming experience. If researchers forget to consider the overall game experience and users' needs, it may result in a game that is difficult to play, confusing, unenjoyable, and even harmful to users. It is essential to balance technical viability and considerations for human factors to ensure the development of successful and user-centered IoT games and GS.

6.2 Is research on IoTGGS exploring the potential and innovation of IoT?

As we mentioned before, the analyzed studies usually focus on technical components of the IoT service structure [46]. However, to achieve IoTGGS's full potential, game design factors and the players' experience should also receive special attention in the development

process because different approaches to using IoT in games require different adaptations of game elements (such as graphics, sounds, narrative, interface, and interaction, among others) [28, 46]. Besides, interaction models and game elements or gamification tactics are crucial for games that use new technologies, as they can enhance the players' acceptance and enjoyment [40]. Thus, researchers and practitioners should consider these factors as core elements to create a sense of immersion and a complex experience (in a good way) with IoTGGS.

Besides that, we can observe a lack of innovation regarding the application domains of the analyzed IoTGGS (see Table 4) when most IoTGGS fell in the same set of categories (Social Life & Entertainment, Smart Environment, Energy Conservation, Health, and Smart Homes), and the entertainment domain encompassed almost 50% of the results. As we said before, this scenario indicates overlooked opportunities to conduct research with IoTGGS exploring varied domains and their impacts on users and their contexts.

Even though our literature research found no "groundbreaking information," these results highlight the limited attention given to human factors and show a still shy search for innovation in IoTGGS development. **Our findings suggest that these technologies have yet to fully harness the potential of IoT and explore innovative interaction modes and game mechanics.** They also reveal a lack of attention given to the design process and HCI evaluation, indicating a missed opportunity to leverage the unique capabilities that IoT promises to deliver. Consequently, **the current state of research on IoTGGS falls short of realizing the envisioned potential and innovation IoT enthusiasts advocate.**

It is worth noting that, given IoTGGS' unique character, particular considerations are necessary when designing and developing such a game/GS [28, 42]. For instance, game design decisions should be supported by real-world understanding and deep comprehension of real people, so the game can fruitfully incorporate the players' contexts, taking into account their preferences and limitations, allowing players to naturally express themselves *through* and *on* the game [18, 40, 46]. By considering users' needs, preferences, and experiences, developers can create games that are not only technologically advanced but also user-friendly, enjoyable, and engaging [66]. We must emphasize the importance of advocating for properly evaluating user interaction and experience to ensure the success of IoTGGS in promoting a human-centric and empowering IoT ecosystem and exploring innovation beyond technical construction.

6.3 Does IOTGGS research pose specific research challenges related to user interaction?

There is a considerable body of literature on works that indicate relevant challenges regarding interactive systems design and evaluation, especially when considering games and gamified interactions – for instance, Darin and colleagues [19] highlight four main challenges when evaluating digital games considering human factors. They are probably not different for IoTGGS since integrating IoT systems into games can be challenging due to the complexity and diversity of IoT technologies, besides the conceptual and human issues [43]. The design of IoTGGS, in theory, requires a multidisciplinary approach that involves experts in HCI, game design, IoT systems, and target domain (e.g., education or health), besides the development of appropriate evaluation methods and metrics.

Notwithstanding, the analyzed papers did not discuss research challenges regarding specific characteristics and requirements of users interacting with IoT devices in gaming interaction. When the authors described any challenges they faced while developing their research, they tended to focus on technical issues (e.g., networking, communication, sensing). In addition, we observed a surprising lack of attention given to ethical issues specific to IoTGGS. Out of the selected studies, a majority failed to mention concerns or measures focused on ethical considerations, even though the ethical issues surrounding IoTGGS are complex and multifaceted. For instance, privacy, security, informed consent, data ownership, algorithmic behavior, and fairness are critical areas that require further attention and integration into the design, evaluation, and implementation of IoTGGS. Even so, most studies failed to acknowledge and discuss such issues.

These results highlight that IoTGGS research has still failed to pose specific challenges to the area and emphasize the pressing need and importance of incorporating human factors and considering HCI principles in the design and evaluation of IoTGGS. That would be important because, by doing so, we could address ethical concerns, enhance the user experience, and maximize the impacts of these innovative gaming applications.

6.4 Moving Forward IoTGGS Research

We now present some insights that can represent possible directions for future research on IoTGGS. We expect to raise reflections that hopefully will motivate new themes, debates, and future work about the design and evaluation of IoTGGS.

6.4.1 Building a Stronger Theoretical Basis. The design of IoTGGS still seems to lack a solid theoretical basis and clarification of concepts. While there are many studies dedicated to examining other innovative types of games (e.g., pervasive and hybrid games), IoTGGS's research and practice still lack a solid theoretical basis and clarification of concepts. IoTGGS research falls behind in clarifying what these games mean and the possibilities they open for designers and players. Most studies are limited to using generic and sometimes confusing terms that indicate a plethora of technology facets but do not transmit the essence of the presented game. Although the inconsistency and disagreement in terms and concepts are common in games studies [4], these indications reflect a yet little-explored domain, suggesting a research gap yet to be developed in future studies on IoTGGS. To fully explore the power of innovative games and avoid misalignment between conceptual and empirical focus on game elements, we need to build a solid knowledge of the area, building on more refined definitions and frameworks to study IoTGGS. We hope this research help to promote these reflections and advance discussions on this subject.

6.4.2 Exploring the Human Side in IoTGGS. IoT scholars envision a complex, intricate relationship between things, people, and environments. A technology-centered perspective can hardly capture these relations without "incurring the risk to reduce the person to a passive subject of the 'smart' technology" [57]. As Soro et al. discussed, there is a lack of a 'humans'-oriented vision in the IoT

perspective [57]. Our results reinforce their conclusions and indicate that games developed under the IoT paradigm would benefit from greater attention to the human aspects of the interaction, at least in the initial phases of the game/GS conceptualization. We believe that if researchers consider human facets as thoroughly as they consider the technology, games developed using IoT would have opportunities to become more autonomous and creative and engage players in more meaningful interactions. In this context, it is essential to highlight the relevance of considering Human Factors, particularly within the field of HCI. By incorporating HCI principles, researchers and practitioners can better understand and address the ethical concerns associated with IoTGGS. This interdisciplinary approach allows for a comprehensive examination of the human-centered aspects of IoTGGS, taking into account user needs, preferences, and ethical considerations. HCI research can provide valuable insights into designing, implementing, and using IoT devices in a way that respects individuals and fosters a positive user experience.

6.4.3 Exploring Specific Ethical Concerns. Some scholars defend that the IoT will radically alter how people interact with technologies [5, 22]. For that reason, it is essential to highlight that IoT also raises concerns and several challenges related to its adoption [22]. The development and use of IoT systems have been seen as a challenge and a complex problem, exacerbated by technical, human, social, and ethical issues. For this reason, despite advances in the development of IoT solutions, studies are still needed to address the diverse aspects linked to this technology and its applications, thoroughly examining its implications for users [8, 35]. Once IoTGGS are part of the IoT paradigm, it seems surprising that such a small portion (18.18%) of the studies mention ethical concerns. We witness a realignment in HCI research, where the interest in ethical design is growing as the purpose of HCI expands from "helping system developers create useful and usable technologies" to a path more attentive to technology's impact on existential questions [37]. Therefore, future research in IoTGGS should promote the conscious design of such applications to guarantee that players feel safe and respected, knowing that the technology is not harming their welfare. Researchers must scrutinize the consequences of the designed technology on users and society.

6.4.4 Empowering IoTGGS Design and Evaluation. The changes resulting from the IoT paradigm emphasize the need for us to rethink some of the conventional approaches used to design and evolve computational solutions and the concepts that base them [10, 45]. This scenario is no different for IoTGGS. Our analysis showed some challenges authors faced when developing the IoTGGS (Section 5). These difficulties indicate the necessity of offering technical and conceptual support to guide and help practitioners and novice researchers working with such systems to produce engaging and innovative games. Given IoTGGS's complexity and the vast range of purposes and application domains they can occupy, there is room for new, diverse research committed to instrumentalizing tools for designing and evaluating them. To address this gap, it is crucial to invest in HCI knowledge and user research, adopting human-centric design practices. By prioritizing user needs, preferences, and experiences, researchers and developers can unlock the true potential of IoT in gaming and foster the creation of engaging, immersive, and transformative IoT game experiences [66]. This call to action encourages researchers and practitioners to embrace human-centered design principles and integrate HCI expertise into the development of IoTGGS, thereby unlocking the innovative possibilities that IoT technology holds.

7 CONCLUSION

This paper presented an SM on IoTGGS, whose main objective was to investigate if and how IoTGGS differ from other innovative games. Our results showed that the main difference between these games is that IoTGGS primarily focus on leveraging the IoT technology, tending to emphasize the technology, while pervasive and hybrid games encompass a broader scope not tied to a particular technology. As contributions, we offered a brief panorama of the field and discussed gaps and indications of possible new directions for IoTGGS research.

This study is not free from limitations common to most SM studies – e.g., the number and nature of search sources, the terms used in the search string, and the nature of the selection and filtering process. A significant limitation refers to our scope: the choice of not searching for studies focused on other types of games (as noted in Section 4). As this was a conscious choice, we are aware that this decision restricts the breadth of the results by excluding some relevant works. However, we believe it helped us to look directly at studies that claim to specifically explore the convergence of IoT and games and identify the presented shortcomings. Our findings are valuable, but deeper studies on this topic are welcomed and needed.

Thus, we suggest, for future work, a deeper comparative analysis of the literature on IoTGGS and pervasive and hybrid games, which could provide valuable insights and a broader understanding of the relations between these domains. Such work would help extend conceptual discussions and build a stronger theoretical basis for IoTGGS. Besides, future research can explore the directions we posed in Section 6.4, for example, by investigating the user interaction considering IoTGGS particularities, ethical issues, and related contexts or proposing guidelines or even new methods and approaches to design and evaluate IoTGGS exploring a human-centered perspective.

After all, the question remains: Is IoTGGS research facing a temporary setback, a "summertime sadness," or was its rise nothing but a "lust for hype"? Our results suggest it is possible that research in this area was impulsed by the "hype" of IoT popularization, as it happens with many new technologies. Maybe this "lust for hype" could explain why few studies use IoT-related terms. Perhaps researchers specialized in HCI games research and games studies are exploring the convergence of IoT and games, but they use more consolidated terms related to umbrella categories, like pervasive or hybrid games. Like Bollen et al. [13], we view scientific knowledge as cumulative, considering each empirical finding as a promissory note rather than a final conclusion. Thus, further investigation is necessary. This paper serves as a call to action and an initial effort to raise reflections and motivate new themes, challenges, debates, and future research on the interaction, evaluation, and design of IoTGGS. Embracing new directions in our practices can elevate IoTGGS research to a more mature level.

Table 4: Summary of the identified IoTGGS

TYPE	APPLICATION	REF	ID	DESCRIPTION	PURPOSE	GENRE	AUDIENCE	STATUS	MATERIALS	PERCEPTION (SENSORS)	COMMUNICATION (NETWORKS)	MIDDLEWARE
Game	**Health and fitness**	[59]	Taoball	A smart IoT ball-shaped device and a smartphone application to provide customized home-based games for muscle training and working memory training.	Health	Mini-game	Elderly	Prototype	Smartphone, Ball	Degree-of-freedom (DOF) Sensors	Wireless sensor networks (WSN)	Not specified
	Smart Environment	[51]	Educational IoT scavenger hunt	A scavenger hunt educational game integrated into smart learn-ing environment based on Moodle platform.	Education	Adventure	General	Launched	Arduino	Environmental, Infrared, RFID, Mobile phone sensors	Internet Protocol for smart objects, RFID and WSN integration	Not specified
		[25]	SEA	Student Engagement Application (SEA) is a data-driven virtual gaming environment to measure student engagement.	Education	Mini-game	Students	Launched	Smartphone	Mobile phone sensors	Low power technologies, Wireless sensor networks	Application specific
		[44]	Fossil Contents	An educational game that takes place in a smart environment (classroom) and helps students to learn science.	Education	Mini-game	Children	Prototype	Smartphone, projector, touch table	Infrared	Not specified	Not specified
		[50]	Educational IoT treasure hunt	An educational, treasure hunt game that aims to increase the interest and motivation of students to study IoT. It is part of a smart learning environment	Education	Adventure	Students	Ideia/Concept	Smartphone, Arduino	Environmental, Infrared, RFID, Mobile phone sensors	Internet Protocol for smart objects, RFID and WSN integration	Not specified
		[21]	Play poles Play canes Beacon boxes Play watch	Physical toys to incentive children to engage in outdoor activities and free play.	Entertainment	Action	Children	Prototype	Diverse (plywood, copper pipe, velcro strap, LED lights)	Environmental sensors	Internet Protocol for smart objects	Application specific
	Social life & entertainment	[26]	Museum game experience	A self-guided museum tour for families and children with interactive game exhibits to promote learning and provoke reflection about the Peranakan culture.	Culture	Adventure	General	Prototype	Laptops, raspeberry pi, and diverse	RFID, Infrared	Wireless sensor networks	Not specified
		[16]	PacMap	A game that requires the player to collect all the cookies virtually positioned across the streets.	Entertainment	Adventure	General	Prototype	Smartphone	Mobile phone sensors (Location sensor)	Wireless sensor networks	Application specific
		[30]	Barbarossa	A role playing game developed to study the recruiting of participants to evaluate pervasive games.	Entertainment	Role-play	General	Launched	Smartphone	Environmental, Mobile phone sensors (Location sensor, camera)	Low power technologies	Not specified
		[40]	Area control	A virtual version of the game Capture the Flag part of a study of architectural analysis of Pervasive IoT games.	Entertainment	Action	General	Ideia/Concept	Smart devices with BLE	Environmental, Mobile phone sensors (Location sensor)	Low power technologies	Application specific (ContextNet)
		[64]	Soccer Robot	An IoT smart arena to play soccer matches with robots, controlled by mobile phone.	Entertainment	Action	Children, Teenagers	Prototype	Smartphone, joystick, 3D-printed field, Arduino, Raspberry Pi	Environmental (photoeletronic), smartphone (camera)	Bluetooth	Not specified
		[31]	Mould Rush	An online multiplayer strategy game played on a landscape shaped by real, living micro-organisms.	Entertainment	Mini-Game	General	Prototype	Smartphone/Computer, Biotic Processing Unit (BPU), Biological material	Environmental sensors	Wireless sensor networks	Application specific/Service oriented
		[23]	GGL	GGL (Gorilla Game Lab) is a high-tech cognitive enrichment for gorillas incorporating IoT and playful interactions.	Entertainment	Mini-game	Gorillas	Prototype	Arduino, Diverse (plywood, acrylic)	Infrared, Mobile phone sensors (camera, microphone)	Wireless sensor networks	Application specific
		[12]	REAL	REAL (Reality-Enhanced AppLied games) is a suite of serious games developed to explore the concept of "reality-enhanced gaming".	Entertainment	Various	General	Prototype	Smartphone/Tablet	Varied for each game	Wireless sensor networks	Service oriented
	Suply chain and logistics	[11]	Beware	A multi-player online game implemented in a mixed space that recreates the environment of a production company.	Education	Adventure, Resource	Students	Prototype	Arduino + living lab	Environmental, RFID, Mobile phone sensors (movement and location)	RFID and WSN integration, Wireless sensor networks	Service oriented
		[60]	BIM-IoT-LC game	A serious game for construction education. It is part of a learning platforms.	Education	Mini-Game	Students	Prototype	Smartphones/Tablets	Mobile phone sensors	NFC, Wireless sensor networks	Service oriented
GS	Energy conservation	[36]	Crowdsourcing Sustainability	A case study for a framework for sensor data analytics that uses gamification to leverage the power of crowdsourcing. It aims to encourage sustainable actions among on users	Behavior Changing	-	General	Prototype	Smartphone	Mobile phone sensors (camera)	Wireless sensor networks	Service/Database oriented
		[39]	ES-CPS	An adaptive and playful cyber-physical system for energy saving to facilitate collaboration between a person and a smart energy-saving system.	Behavior Changing	-	General	Prototype	Domestic ppliances (TVs, laptops, lights)	Infrared, Environmental sensors	Low power technologies, Wireless sensor networks	Application specific
		[20]	ChArGED	ChArGED is part a gamified framework that aims to change occupants' energy-consumption behaviors and reduce energy wastage in public buildings.	Behavior Changing	-	General	Prototype	Smartphone	Environmental sensors	NFC, Low power technologies, Wireless sensor networks	Event based
	Smart Homes	[63]	Smart Home Assignment System	A home management system with gamification for small tasks. The system supervise assigned tasks and give points when a task is done.	Behavior Changing	-	Families, roommates	Prototype	Smartphone	Environmental, Mobile phone sensors (Camera, movement sensor)	Not specified	Database oriented
		[48]	Solis	A gamified plant care system that encourages people to take care of their houseplants.	Behavior Changing	-	General	Prototype	Smartphone	Environmental sensors	Wireless sensor networks	Service oriented
	Social life & entertainment	[38]	Gamified Smart Objects for Museums	A trivia quiz game and gamified smart object platform to automate the creation of questions about exhibits.	Culture	-	General	Launched	Smartphone	Mobile phone sensors (camera)	Wireless sensor networks	Service oriented

REFERENCES

[1] Shabir Ahmad, Faisal Mehmood, Faheem Khan, and Taeg Keun Whangbo. 2022. Architecting Intelligent Smart Serious Games for Healthcare Applications: A Technical Perspective. *Sensors* 22, 3 (2022), 810.

[2] Tanveer Ahmad and Dongdong Zhang. 2021. Using the internet of things in smart energy systems and networks. *Sustainable Cities and Society* 68 (2021), 102783.

[3] Fahed Alkhabbas, Romina Spalazzese, and Paul Davidsson. 2019. Characterizing Internet of Things systems through taxonomies: A systematic mapping study. *Internet of Things* 7 (2019), 100084.

[4] Abdelhadi Alla and Khalid Nafil. 2019. Gamification in IoT application: A systematic mapping study. *Procedia Computer Science* 151 (2019), 455–462.

[5] Fritz Allhoff and Adam Henschke. 2018. The internet of things: Foundational ethical issues. *Internet of Things* 1 (2018), 55–66.

[6] Jeferson Arango-López, Luis Fernando Castillo Ossa, Cesar Alberto Collazos, Francisco Luis Gutiérrez Vela, and Habib M Fardoun. 2019. Pervasive games in the rehabilitation process: a systematic literature review. In *Proceedings of the 5th Workshop on ICTs for improving Patients Rehabilitation Research Techniques*. 140–143.

[7] Jeferson Arango-López, Cesar A Collazos, Francisco Luis Gutiérrez Vela, and Luis F Castillo. 2017. A systematic review of geolocated pervasive games: a perspective from game development methodologies, software metrics and linked open data. In *Design, User Experience, and Usability: Designing Pleasurable Experiences: 6th International Conference, DUXU 2017, Held as Part of HCI International 2017, Vancouver, BC, Canada, July 9-14, 2017, Proceedings, Part II 6*. Springer, 335–346.

[8] Parvaneh Asghari, Amir Masoud Rahmani, and Hamid Haj Seyyed Javadi. 2019. Internet of Things applications: A systematic review. *Computer Networks* 148 (2019), 241–261.

[9] Luigi Atzori, Antonio Iera, and Giacomo Morabito. 2010. The internet of things: A survey. *Computer networks* 54, 15 (2010), 2787–2805.

[10] Luigi Atzori, Antonio Iera, and Giacomo Morabito. 2017. Understanding the Internet of Things: definition, potentials, and societal role of a fast evolving paradigm. *Ad Hoc Networks* 56 (2017), 122–140.

[11] Jannicke Madeleine Baalsrud Hauge, Theodore Lim, Matthias Kalverkamp, Florian Haase, and Francesco Bellotti. 2016. Analysis on educating mechanical engineers through serious games using pervasive technologies. In *International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, Vol. 50084. American Society of Mechanical Engineers, V01BT02A050.

[12] Francesco Bellotti, Riccardo Berta, Pratheep Paranthaman, Gautam Dange, and Alessandro De Gloria. 2019. REAL: Reality-enhanced applied games. *IEEE Transactions on Games* 12, 3 (2019), 281–290.

[13] Kenneth Bollen, JT Cacioppo, RM Kaplan, JA Krosnick, and JL Olds. 2015. Social, behavioral, and economic sciences perspectives on robust and reliable science: Report of the Subcommittee on Replicability in Science, Advisory Committee to the National Science Foundation Directorate for Social, Behavioral, and Economic Sciences. *National Science Foundation* (2015).

[14] Paul Brous, Marijn Janssen, and Paulien Herder. 2020. The dual effects of the Internet of Things (IoT): A systematic review of the benefits and risks of IoT adoption by organizations. *International Journal of Information Management* 51 (2020), 101952.

[15] Nayana Carneiro, Ticianne Darin, and Windson Viana. 2019. What are we talking about when we talk about location-based games evaluation? a systematic mapping study. In *Proceedings of the 18th Brazilian Symposium on Human Factors in Computing Systems*. 1–13.

[16] Thomas Chatzidimitris, Damianos Gavalas, and Vlasios Kasapakis. 2014. PacMap: transferring PacMan to the physical realm. In *International Internet of Things Summit*. Springer, 139–144.

[17] Ioannis Chatzigiannakis and Christos Tselios. 2021. Internet of Everything. In *Intelligent Computing for Interactive System Design: Statistics, Digital Signal Processing, and Machine Learning in Practice*. 21–56.

[18] Liliana Vale Costa, Ana Isabel Veloso, and Óscar Mealha. 2017. A review of proxemics in 'smart game-playing'. In *Conference on Smart Learning Ecosystems and Regional Development*. Springer, 219–226.

[19] Ticianne Darin, Nayana Carneiro, David Miranda, and Bianca Coelho. 2023. Challenges in Evaluating Players' Interaction with Digital Games. In *Grand Research Challenges in Games and Entertainment Computing in Brazil-GranDGamesBR 2020–2030: First Forum, GranDGamesBR 2020, Recife, Brazil, November 7-10, 2020, and Second Forum, GranDGamesBR 2021, Gramado, Brazil, October 18–21, 2021, Revised Selected Papers*. Springer, 1–24.

[20] Nikos Dimitriou, Anastasia Garbi, Kostas Vasilakis, Anthony Schoofs, Amr Taha, Manolis Nikiforakis, Sarantis Kotsilitis, Thanasis G Papaioannou, Dimosthenis Kotsopoulos, Cleopatra Bardaki, et al. 2018. ChArGED: Implementing a framework for improving energy efficiency in public buildings through IoTenabled energy disaggregation and serious games. In *2018 IEEE International Conference on Pervasive Computing and Communications Workshops (PerCom Workshops)*. IEEE, 65–70.

[21] Thomas Dylan, Gavin Wood, Abigail C Durrant, John Vines, Pablo E Torres, Philip IN Ulrich, Mutlu Cukurova, Amanda Carr, Sena Çerçi, and Shaun Lawson. 2020. Designing IoT Resources to Support Outdoor Play for Children. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–12.

[22] David Fletcher. 2015. Internet of things. In *Evolution of cyber technologies and operations to 2035*. Springer, 19–32.

[23] Stuart Gray, Fay Clark, Katy Burgess, Tom Metcalfe, Anja Kadijevic, Kirsten Cater, and Peter Bennett. 2018. Gorilla Game Lab: Exploring modularity, tangibility and playful engagement in cognitive enrichment design. In *Proceedings of the Fifth International Conference on Animal-Computer Interaction*. 1–13.

[24] Stephanie Heintz and Effie Lai-Chong Law. 2015. The game genre map: A revised game classification. In *Proceedings of the 2015 Annual Symposium on Computer-Human Interaction in Play*. 175–184.

[25] John Henry, Stephen Tang, Subhas Mukhopadhyay, and Moi Hoon Yap. 2021. A randomised control trial for measuring student engagement through the Internet of Things and serious games. *Internet of Things* 13 (2021), 100332.

[26] Hai Huang and Kher Hui Ng. 2021. Designing for cultural learning and reflection using IoT serious game approach. *Personal and Ubiquitous Computing* 25, 3 (2021), 509–524.

[27] Helene Joffe. 2012. Thematic analysis. *Qualitative research methods in mental health and psychotherapy* 1 (2012), 210–223.

[28] Petter Bakkan Johansen. 2018. *Iot-based pervasive game framework-a proof of concept case study*. Master's thesis. NTNU.

[29] Vlasios Kasapakis and Damianos Gavalas. 2015. Pervasive gaming: Status, trends and design principles. *Journal of Network and Computer Applications* 55 (2015), 213–236.

[30] Vlasios Kasapakis, Damianos Gavalas, and Thomas Chatzidimitris. 2014. Evaluation of pervasive games: recruitment of qualified participants through preparatory game phases. In *International Internet of Things Summit*. Springer, 118–124.

[31] Raphael Kim, Siobhan Thomas, Roland van Dierendonck, and Stefan Poslad. 2018. A new mould rush: designing for a slow bio-digital game driven by living microorganisms. In *Proceedings of the 13th International Conference on the Foundations of Digital Games*. 1–9.

[32] Barbara Kitchenham, O Pearl Brereton, David Budgen, Mark Turner, John Bailey, and Stephen Linkman. 2009. Systematic literature reviews in software engineering–a systematic literature review. *Information and software technology* 51, 1 (2009), 7–15.

[33] Barbara A Kitchenham, David Budgen, and O Pearl Brereton. 2010. The value of mapping studies-A participant-observer case study.. In *EASE*, Vol. 10. 25–33.

[34] Mehmet Kosa and Pieter Spronck. 2018. What tabletop players think about augmented tabletop games: a content analysis. In *Proceedings of the 13th International Conference on the Foundations of Digital Games*. 1–8.

[35] Aleksandr Lepekhin, Alexandra Borremans, Igor Ilin, and Sami Jantunen. 2019. A systematic mapping study on internet of things challenges. In *2019 IEEE/ACM 1st International Workshop on Software Engineering Research & Practices for the Internet of Things*. IEEE, 9–16.

[36] Alexandra L'Heureux, Katarina Grolinger, Wilson A Higashino, and Miriam AM Capretz. 2017. A gamification framework for sensor data analytics. In *2017 IEEE international congress on internet of things (ICIOT)*. IEEE, 74–81.

[37] Sharon Lindberg, Petter Karlström, and Sirkku Männikkö Barbutiu. 2021. Design Ethics in Practice-Points of Departure. *Proceedings of the ACM on Human-Computer Interaction* 5, CSCW1 (2021), 1–19.

[38] Alejandro López-Martínez, Carlos A Iglesias, and Álvaro Carrera. 2020. Gamified Smart Objects for Museums Based on Automatically Generated Quizzes Exploting Linked Data. In *2020 16th International Conference on Intelligent Environments (IE)*. IEEE, 132–139.

[39] Ching-Hu Lu. 2018. IoT-enabled adaptive context-aware and playful cyber-physical system for everyday energy savings. *IEEE Transactions on Human-Machine Systems* 48, 4 (2018), 380–391.

[40] André MacDowell and Markus Endler. 2014. Internet of things based multiplayer pervasive games: An architectural analysis. In *International Internet of Things Summit*. Springer, 125–138.

[41] Carsten Magerkurth. 2012. Hybrid gaming environments: keeping the human in the loop within the Internet of things. *Universal Access in the Information Society* 11, 3 (2012), 273–283.

[42] Ahsan Manzoor, Max Samarin, David Mason, and Mika Ylianttila. 2020. Scavenger Hunt: Utilization of Blockchain and IoT for a location-based Game. *IEEE Access* 8 (2020), 204863–204879.

[43] M. Mendes, A. Pereira, L. Almeida, and J. Barata. 2021. Challenges and Opportunities on Using Games to Support IoT Systems Teaching. In *2021 IEEE Global Engineering Education Conference (EDUCON)*. 1–8. https://doi.org/10.1109/EDUCON46848.2021.9513484

[44] Lee Jeong Min, Kim Ji Hye, and Lee Kyoung Hak. 2016. Development of Elementary Science Fossil Learning. *Indian Journal of Science and Technology* 9 (2016), 46.

[45] Daniele Miorandi, Sabrina Sicari, Francesco De Pellegrini, and Imrich Chlamtac. 2012. Internet of things: Vision, applications and research challenges. *Ad hoc networks* 10, 7 (2012), 1497–1516.

[46] Lee MyounJae. 2016. Secure game development for IoT environment. *Journal of Computer Virology and Hacking Techniques* 12, 3 (2016), 125–130.

[47] Felipe Paiva, Glaudiney Mendonça, and Windson Viana. 2022. A Systematic Mapping of Hybrid Games in the Academy. In *Anais Estendidos do XXI Simpósio Brasileiro de Jogos e Entretenimento Digital*. SBC, 128–137.

[48] Arno Penders, Johanna Renny Octavia, Michiel Caron, Fay de Haan, Thomas Devoogdt, Samnang Nop, Anthony McAtear, Olivier Pieters, Francis Wyffels, Steven Verstockt, et al. 2018. Solis: A smart interactive system for houseplants caring. In *2018 International Conference on Orange Technologies (ICOT)*. IEEE, 1–7.

[49] Kai Petersen, Sairam Vakkalanka, and Ludwik Kuzniarz. 2015. Guidelines for conducting systematic mapping studies in software engineering: An update. *Information and Software Technology* 64 (2015), 1–18.

[50] Luka Petrović, Danijela Stojanović, Aleksandra Labus, Zorica Bogdanović, and Marijana Despotović-Zrakić. 2017. Harnessing Edutainment in Higher Education: an example of an IoT based game. In *The 12th International conference on virtual learning, Sibiu, Romania-Europe*. 318–324.

[51] Luka Petrović, Danijela Stojanović, Svetlana Mitrović, Dušan Barać, and Zorica Bogdanović. 2022. Designing an extended smart classroom: An approach to game-based learning for IoT. *Computer applications in engineering education* 30, 1 (2022), 117–132.

[52] Partha Pratim Ray. 2018. A survey on Internet of Things architectures. *Journal of King Saud University-Computer and Information Sciences* 30, 3 (2018), 291–319.

[53] Gianna Reggio, Maurizio Leotta, Maura Cerioli, Romina Spalazzese, and Fahed Alkhabbas. 2020. What are IoT systems for real? An experts' survey on software engineering aspects. *Internet of Things* 12 (2020), 100313.

[54] James Scheibner, Anna Jobin, and Effy Vayena. 2021. Ethical issues with using Internet of Things devices in citizen science research: A scoping review. *Frontiers in Environmental Science* 9 (2021), 629649.

[55] Ben Schouten, Gabriele Ferri, Michiel de Lange, and Karel Millenaar. 2017. Games as strong concepts for city-making. In *Playable cities*. Springer, 23–45.

[56] Pallavi Sethi and Smruti R Sarangi. 2017. Internet of things: architectures, protocols, and applications. *Journal of Electrical and Computer Engineering* 2017 (2017).

[57] Alessandro Soro, Margot Brereton, Paul Roe, Peta Wyeth, Daniel Johnson, Aloha Hufana Ambe, Ann Morrison, Shaowen Bardzell, Tuck Wah Leong, Wendy Ju, et al. 2017. Designing the social internet of things. In *Proceedings of the 2017 CHI Conference Extended Abstracts on Human Factors in Computing Systems*. 617–623.

[58] Benjamin Stokes, Hazel Arroyo, Mitchell Loewen, Tambra Stevenson, and Chris J Karr. 2020. A Playful City in the Cards: Sharing Power in Game Design by Extending the Card Metaphor. In *Extended Abstracts of the 2020 Annual Symposium on Computer-Human Interaction in Play*. 375–378.

[59] Chen-Yu Su, Chun-Yi Hsiao, Ren-Guey Lee, and Jyh-Horng Chen. 2016. TaoBall: An interactive IoT ball design for rehabilitation. In *2016 IEEE 6th International Conference on Consumer Electronics-Berlin (ICCE-Berlin)*. IEEE, 24–27.

[60] Jochen Teizer, Olga Golovina, Stephan Embers, and Mario Wolf. 2020. A serious gaming approach to integrate BIM, IoT, and lean construction in construction education. In *Construction Research Congress 2020: Project Management and Controls, Materials, and Contracts*. American Society of Civil Engineers Reston, VA, 21–30.

[61] Jinke D Van Der Laan, Adriaan Heino, and Dick De Waard. 1997. A simple procedure for the assessment of acceptance of advanced transport telematics. *Transportation Research Part C: Emerging Technologies* 5, 1 (1997), 1–10.

[62] Effy Vayena. 2021. Ethical issues with using Internet of Things devices in citizen science research: a scoping review. *Journal of Medical Ethics* 60, 2 (2021), 247.

[63] Alicja Winnicka, Karolina Kęsik, Dawid Połap, Marcin Woźniak, and Zbigniew Marszałek. 2019. A multi-agent gamification system for managing smart homes. *Sensors* 19, 5 (2019), 1249.

[64] Bin Wu, Zhen Yang, Xinyang Wu, and Yuqing Chen. 2020. 3D Printing Intelligent Soccer Robot Match System. In *2020 Chinese Automation Congress (CAC)*. IEEE, 6013–6017.

[65] Ruowei Xiao, Zhanwei Wu, and Juho Hamari. 2021. Internet-of-Gamification: A Review of Literature on IoT-enabled Gamification for User Engagement. *International Journal of Human–Computer Interaction* (2021), 1–25.

[66] Kaja Fjørtoft Ystgaard, Luigi Atzori, David Palma, Poul Einar Heegaard, Lene Elisabeth Bertheussen, Magnus Rom Jensen, and Katrien De Moor. 2023. Review of the theory, principles, and design requirements of human-centric Internet of Things (IoT). *Journal of Ambient Intelligence and Humanized Computing* 14, 3 (2023), 2827–2859.